ROSHN Deployment Requirements

Overview

This document summarizes the requirements for deploying the ROSHN Parametric Masterplan Modelling Platform. The platform supports three deployment options: Azure Cloud, Google Cloud Platform (GCP), and On-Premise servers.

Current Status

Item	Status
PoC Completion	Completed (Azure Cloud, UAE)
Business Validation	Processes and use cases confirmed
Production Deployment	Pending (4-8 weeks from kick-off)

Technical Documentation

All technical documentation is available for ROSHN IT and GRC review:

Architecture Documents

Document	Description	Link
High-Level Design (HLD)	System architecture, technology stack, security layers	View HLD
Detailed Design (DLD)	Database schema, API specs, authentication flows	View DLD
Infrastructure Sizing	Server specifications for all environments	View Sizing

Deployment Guides

Document	Description	Link
Azure Deployment	Complete Azure deployment guide	View Guide
GCP Deployment	Complete GCP deployment guide	View Guide
On-Premise Deployment	Complete on-premise deployment guide	View Guide

Calculation Documentation

Document	Description	Link
Calculations Overview	All calculation formulas and data flow	View Calculations
Building Assets	Building cost calculation methodology	View Doc
Car Parking	Parking cost calculation methodology	View Doc

Infrastructure	Infrastructure cost calculation	View Doc
Public Realm	Public realm cost calculation	View Doc

Deployment Options

Option 1: Microsoft Azure

Best for organizations already using Azure or requiring UAE-based hosting.

Environment	App Service	Database	Use Case
dev	B1 (Basic)	B_Gen5_1 (Basic, 1 vCore)	Development and testing
staging	S1 (Standard)	GP_Gen5_2 (2 vCores)	Pre-production testing
prod	P1V2 (Premium)	GP_Gen5_4 (4 vCores)	Production workloads

Deployment Command:

```
./scripts/deploy-azure.sh [dev|staging|prod]
```

Region: UAE North (uaenorth)

Full Guide: Azure Deployment

Option 2: Google Cloud Platform (GCP)

Best for GCP Dammam region deployment (Saudi Arabia data residency).

Environment	Cloud SQL	Cloud Run	Scaling	Use Case
dev	db-f1-micro	512Mi, 1 CPU	0-2 instances	Development (scale to zero)
staging	db-custom-1-3840	1Gi, 1 CPU	0-5 instances	Pre-production testing
prod	db-custom-2-4096	2Gi, 2 CPU	1-10 instances	Production (always-on)

Deployment Command:

```
./scripts/deploy-gcp.sh [dev|staging|prod]
```

Region: me-central1 (Dammam, Saudi Arabia)

Full Guide: GCP Deployment

Option 3: On-Premise Servers

Best for full data sovereignty and ROSHN infrastructure integration.

Environment	Port	PM2 Instances	SSL	Domain
dev	3001	1	Optional	dev.platform.roshn.sa
staging	3002	2	Required	staging.platform.roshn.sa
prod	3000	max (all CPUs)	Required	platform.roshn.sa

Deployment Command:

```
sudo ./scripts/deploy-onprem.sh [dev|staging|prod]
```

Full Guide: On-Premise Deployment

Infrastructure Requirements

Full Details: Infrastructure Sizing Document

Summary by Environment

Component	Development	Staging	Production
CPU	4 vCPUs	4 vCPUs	8 vCPUs × 2
RAM	8 GB	8 GB	16 GB × 2
Storage	100 GB SSD	150 GB SSD	250 GB SSD × 2
Database	Same server	Separate (2 vCPU, 4GB)	Dedicated (4 vCPU, 16GB)
SSL	Optional	Required	Required
Load Balancer	No	Optional	Required

Deployment Timeline (4-8 Weeks)

Phase 1: Project Initiation

Action	Owner	Status	Documentation
Initiate new demand for project with PM	Ahmad Kawakbi	Pending	-
Share infrastructure sizing details	Omnium/Innovaite	✅ Complete	Infrastructure Sizing
Sign NDA and AUA documents	Omnium team members	Pending	-

Phase 2: Environment Setup

Action	Owner	Status	Documentation

Prepare ROSHN environment (on-prem or GCP)	ROSHN IT	Pending	Infrastructure Sizing
Share technical architecture (HLD)	Omnium/Innovaite	✅ Complete	High-Level Design
Share technical architecture (DLD)	Omnium/Innovaite	✅ Complete	Detailed Design
Align with GRC requirements	Both parties	✅ Docs Ready	HLD + DLD

Phase 3: Deployment & Integration

Action	Owner	Status	Documentation
Onboard vendor on VDI/VPN with PAM	ROSHN IT	Pending	-
Deploy application	Omnium/Innovaite	✅ Scripts Ready	Azure / GCP / On-Prem
SSO integration with ROSHN-IAM	Both parties	✅ Docs Ready	DLD Section 3

Phase 4: Security & Testing

Action	Owner	Status	Documentation
Conduct VAPT post UAT setup	ROSHN GRC	✅ Docs Ready	HLD Section 7 + DLD Section 6
Revise GRC checklist with evidences	Both parties	✅ Docs Ready	All architecture docs
Complete UAT	ROSHN Business	Pending	-

Phase 5: Go-Live

Action	Owner	Status	Documentation
Align with change management processes	Both parties	Pending	-
GRC approval	ROSHN GRC	Pending	-
Production deployment	Omnium/Innovaite	✅ Scripts Ready	Azure / GCP / On-Prem
Production launch	Both parties	Pending	-

Documentation Status

Document	Status	For Phase

[High-Level Design (HLD)]	✅ Complete	Phase 2, 4
[Detailed Design (DLD)]	✅ Complete	Phase 2, 3, 4
[Infrastructure Sizing]	✅ Complete	Phase 1, 2
[Azure Deployment Guide]	✅ Complete	Phase 3, 5
[GCP Deployment Guide]	✅ Complete	Phase 3, 5
[On-Premise Deployment Guide]	✅ Complete	Phase 3, 5

Security & Compliance

Technical Security Documentation

Topic	Document	Section
Security Architecture	HLD	Section 7
Authentication Flow	DLD	Section 3
Password Security	DLD	Section 6.1
Session Management	DLD	Section 6.2
2FA Implementation	DLD	Section 6.3
Audit Logging	DLD	Section 9
Network Security	Infrastructure Sizing	Section 5

Completed Documents

Document	Status
Service Activation Checklist	Submitted
3rd Party Controls Assessment	Submitted

Security Requirements

- VDI/VPN access with PAM (Privileged Access Management)
- SSO integration with ROSHN-IAM
- VAPT (Vulnerability Assessment and Penetration Testing) post-UAT
- GRC checklist compliance with evidences

Post-Production Support

SLA Requirements

Priority	Response Time	Resolution Time
P1 (Critical)	30 minutes	1 hour

P2 (High)	30 minutes	4 hours
P3 (Medium)	1 hour	8 hours
P4 (Low)	4 hours	24 hours

Support Process

- Incident management process applies
- Support via ROSHN IT ticketing system

Key Contacts

ROSHN

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Omnium/Innovaite

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Quick Links

All Documentation

- High-Level Design (HLD)
- Detailed Design (DLD)
- Infrastructure Sizing
- Azure Deployment
- GCP Deployment
- On-Premise Deployment

Deployment Scripts

```
# Clone repository
git clone https://github.com/tajalagawani/roshn.git
cd roshn

# Azure
./scripts/deploy-azure.sh [dev|staging|prod]

# GCP
./scripts/deploy-gcp.sh [dev|staging|prod]

# On-Premise
sudo ./scripts/deploy-onprem.sh [dev|staging|prod]
```